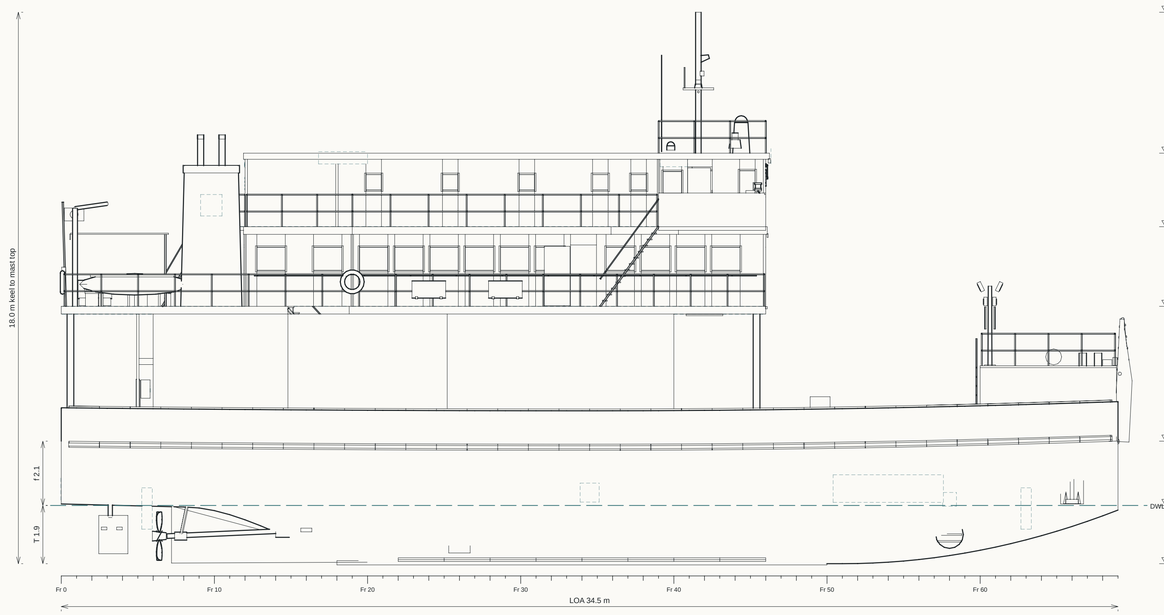
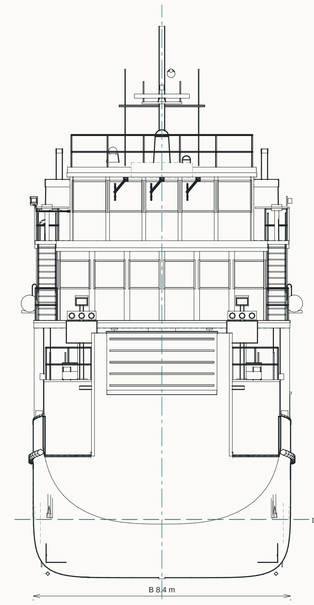
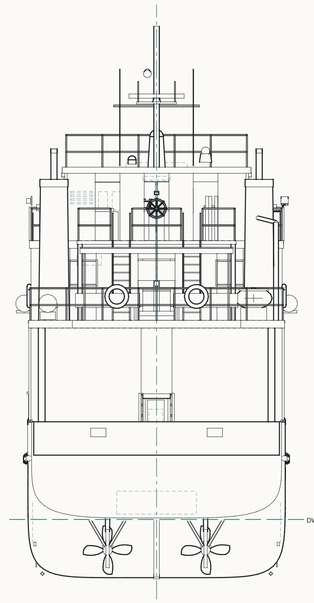


STARBOARD PROFILE · ramp raised, at sea SCALE 1:100



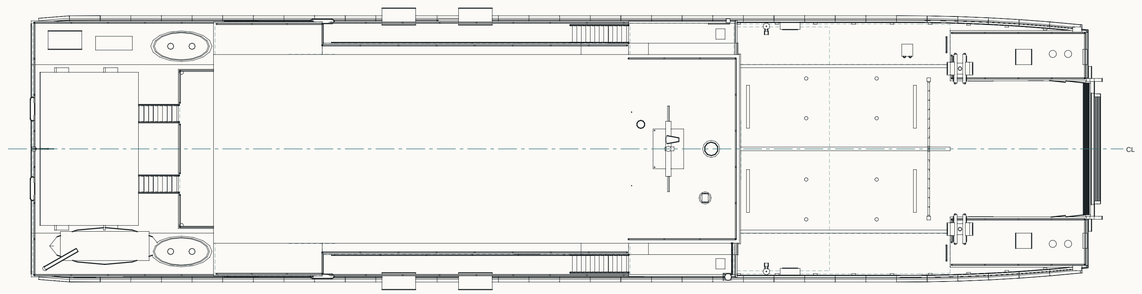
BOW SCALE 1:100

looking aft · port on the right

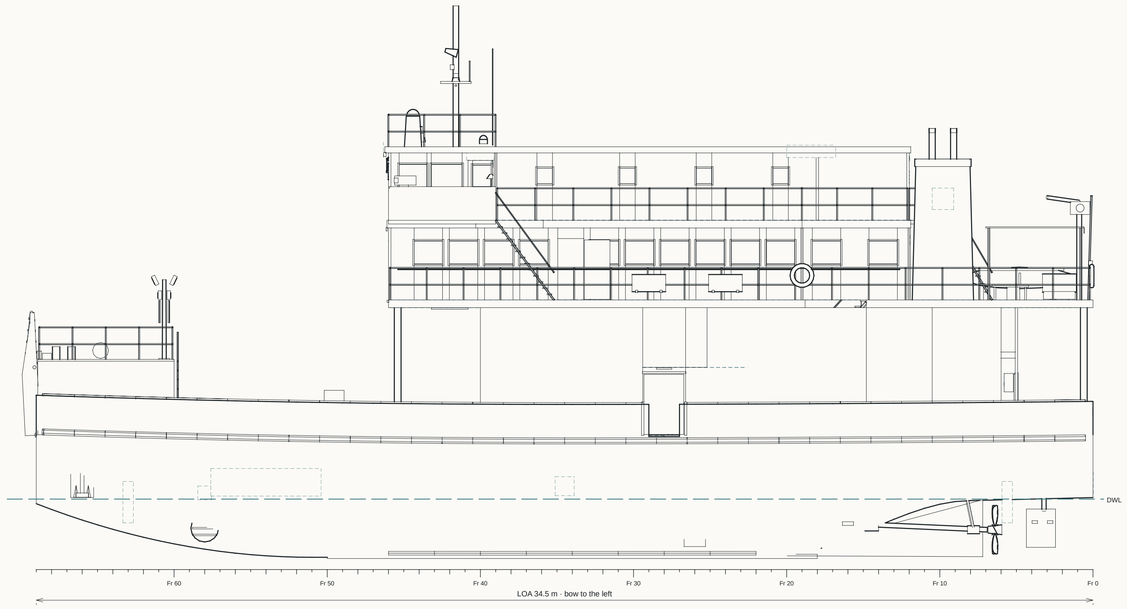


STERN SCALE 1:100

looking forward · starboard on the right



PLAN FROM ABOVE · port side up SCALE 1:100



PORT PROFILE · gangway door and gate in her side SCALE 1:100

PRINCIPAL PARTICULARS

Length overall	34.50 m
Length between perpendiculars	31.00 m
Breadth moulded	8.40 m
Depth to car deck	4.00 m
Draft, laden	1.90 m
Freeboard to car deck	2.10 m
Car deck / passenger deck	4.0 / 8.4 m ABL
Bridge deck / compass deck	11.0 / 13.4 m ABL
Mast top in the data / as modelled	18.0 / 18.00 m ABL
Bow ramp	leaf 4 m, flap 2 m, 4.6 m over girders
Model extent, x (port +)	-4.59 to 4.59 m
Model extent, z (fwd +)	-17.28 to 17.81 m

LINES

- Visible outline: silhouette and free edges
- Visible feature edges (faces meeting at 50° or more)
- Paint: names, draft and load-line marks, red walkways
- Designed waterline, laden 1.9 m
- Centreline

NOTES

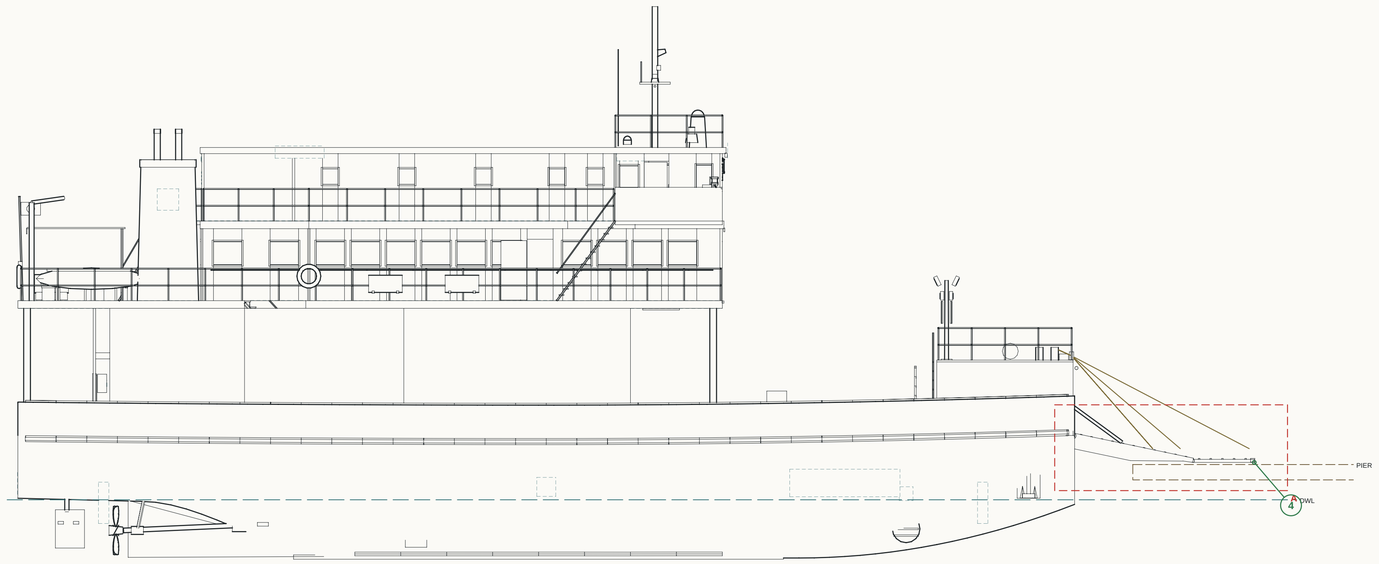
Drawn from the game itself, not redrawn by hand: buildFerryModel() (*src/world/ferry-model.js*) is built in Node exactly as the game builds it; every visible triangle is projected orthographically and hidden lines are removed against a 10 mm depth buffer. What this sheet shows is what is modelled, including its faults.

Heights ABL are above the base line (keel); the model frame is the laden waterline. Frames (Fr) at 500 mm from the transom. Glass is drawn opaque, as an elevation does. The people, mooring lines and the shore are not part of the ship and are not drawn here: see sheet X2.

The DXF beside this sheet is the same drawing in millimetres at sheet size (layers OUTLINE, DETAIL, MARKING, WATERLINE, CENTRE, DIM, TEXT).

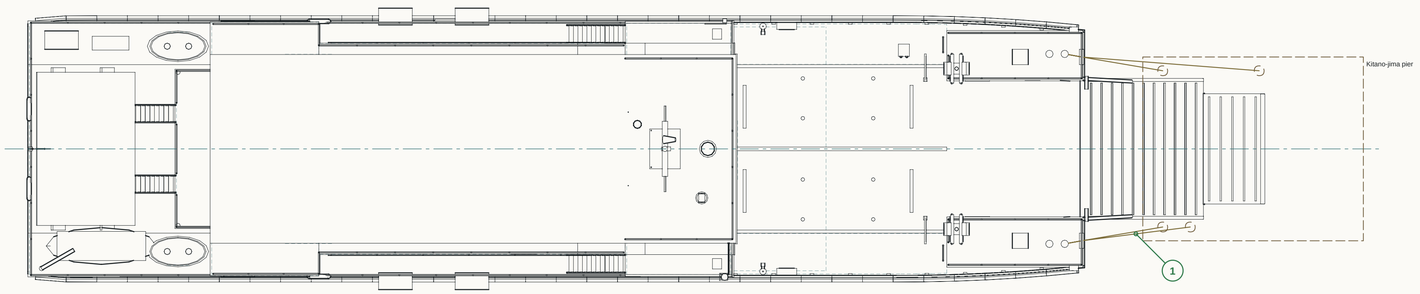
YARD Johansson Shipyard, Innoshima **JOHANSSON SHIPYARD**	SHIP M/V MINATO MARU hull No. 127, built 1994		DRAWING No. **JS-127-X1**
TITLE EXTERIOR AS BUILT · at sea, ramp raised **Profiles, plan, bow and stern from the game model**			SCALE 1:100 on A1
SOURCE tools/ferry-exterior-cad.mjs	DATA buildFerryModel() as built	STATUS AS-BUILT SURVEY	SHEET 1 / 3
DRAWN Claude, AI survey	CHECKED owner review open	DATE 2026-11-03	REV B · fixes, stair, saloon

AS-BUILT SURVEY OF THE GAME MODEL · Johansson World vessel · no flag, class or construction approval · drawn by projecting buildFerryModel() with hidden lines removed



STARBOARD PROFILE · alongside Kitano-jima, ramp on the pier

SCALE 1:100



PLAN · lines as rigged at Kitano-jima

SCALE 1:100

FINDINGS · Rev A found them, Rev B fixed them; measured again from the game model

1 — **At Kitano-jima the after spring and stern line ran through the ship** — RESOLVED

Rev A: the aft station's two lines were led from the port-quarter fairlead to pier bitts ahead of the bow: 37.88 and 37 m through the aft mooring station, casing P, the lower lobby and across the car deck (10 and 9 modelled faces), the yellow line in the screenshot. The Minato head line also bent in the air over the port bow house's roof edge, clipping it.
Rev B: each station keeps its own two ropes (ferry.js FERRY_MOORING.ropes); bow-on at Kitano-jima she makes fast from both bows only, each led through its stern fairlead to the bitts on its own side, the bosun at the starboard bow, the stern ropes coiled aft; Minato's head line and forward spring lead through a new side chock on the bow house's corner; the coils moved clear of the windlass drums.
Now: 0 modelled faces pierced by any laid line at Kitano-jima or Minato, ramp raised or laid (tests/ferry-exterior.test.mjs).

2 — **The bridge name board's two names ran into each other** — RESOLVED

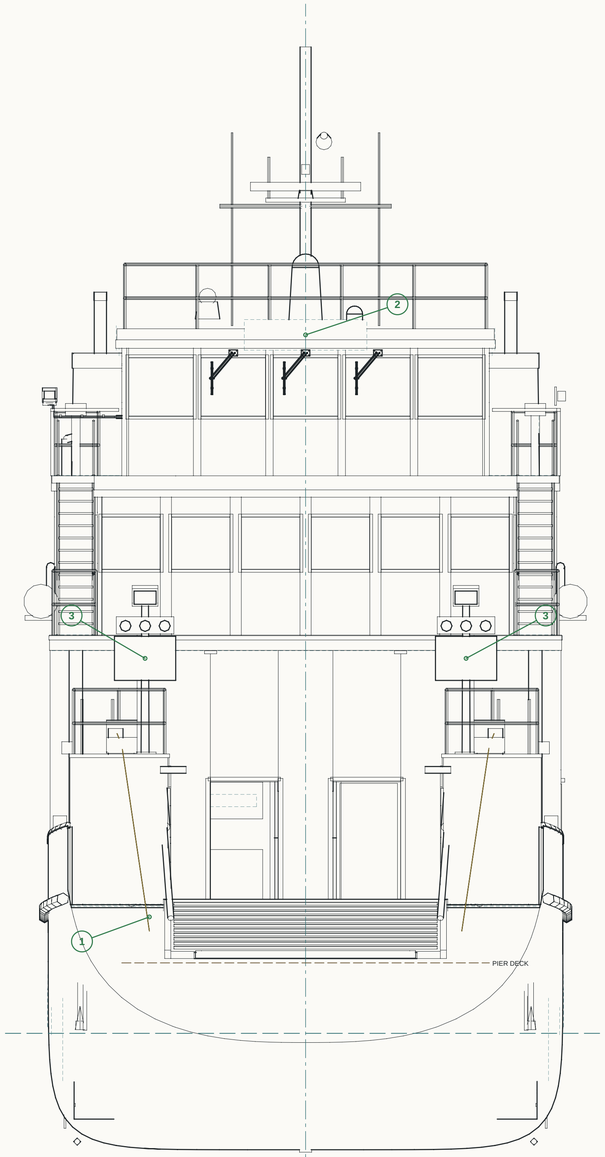
Rev A: みなと丸 and MINATO MARU were each centred on fixed points of the 512 px board (30% and 72%); in Helvetica Bold (an iPad) the English started at 238.7 px and 丸 ended at 263.5 px, about 25 px of overlap.
Rev B: the names are laid out from their measured widths, centred as a pair with 30 px between them inside a 34 px margin, both shrinking together if a device's font is wider (src/world/ferry-boards.js nameBoardLayout).
Now: 26.3 px between the names in Chromium's fonts, 26.1 px in Helvetica Bold.

3 — **The lane boards' truck sign sat across a lane divider** — RESOLVED

Rev A: the truck pictogram was drawn beside the middle arrow's foot, its cab (x 326–348 px) across the divider (329–335 px), 0.18 m tall: from the quay it read as a stray "L" on the arrow. The dividers were 7 px off the middle between the lanes.
Rev B: a larger truck under a shortened centre arrow, centred in its lane; the dividers midway between the lanes (src/world/ferry-boards.js laneBoardLayout).
Now: the truck spans x 198–314 px, 22 and 22 px clear of the dividers.

4 — **The ramp's side girders ran into the pier** — RESOLVED

Rev A: the leaf's two side girders, 0.5 m deep for its whole 4 m and square at the toe, ended 0.28 m inside the Kitano-jima pier deck and 0.33 m below Minato's landing plate; the flap's hinge pin sank 0.02 m into the landing.
Rev B: the girders taper over the leaf's last 2 m to the deck plate's own depth at the toe, the plate's underside eases up over its last 0.3 m, the flap's hinge pin is flush with its underside, and the plate is 4.4 m clear between the girders.
Now: lowest point of the laid ramp 0.000 m over the Kitano-jima pier, 0.000 m over Minato's landing plate.



BOW ELEVATION · as in the screenshot

SCALE 1:50

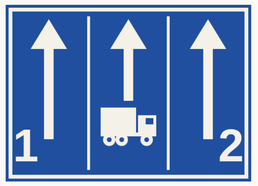
looking aft from the pier · port on the right · lines as rigged

AS PAINTED

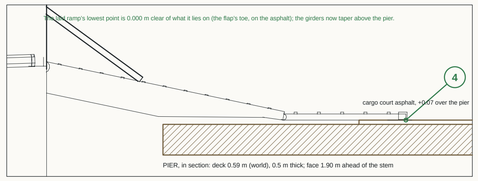
by the game's own painters (src/world/ferry-boards.js)



Name board (bridge front): 26.3 px between the names here, 26.1 px in Helvetica Bold (an iPad). Finding 2.



Lane board, face to the quay: the truck spans 198–314 px between the dividers at 170–176 and 336–342 px. Finding 3.



DETAIL A · the ramp's toe on the pier

SCALE 1:50

NOTES

Her pose is the game's: on her waterline at rest, the ramp laid on the Kitano-jima pier by restRamp(), the lines laid bitt → fairlead → pier bitt as createFerryRun lays them (FERRY_MOORINGS.airport). Green: found in Rev A and fixed, measured again. Red: still open. Red dashed: a faulty line where it is hidden inside or behind her; × where a line pierces a modelled face (none now).

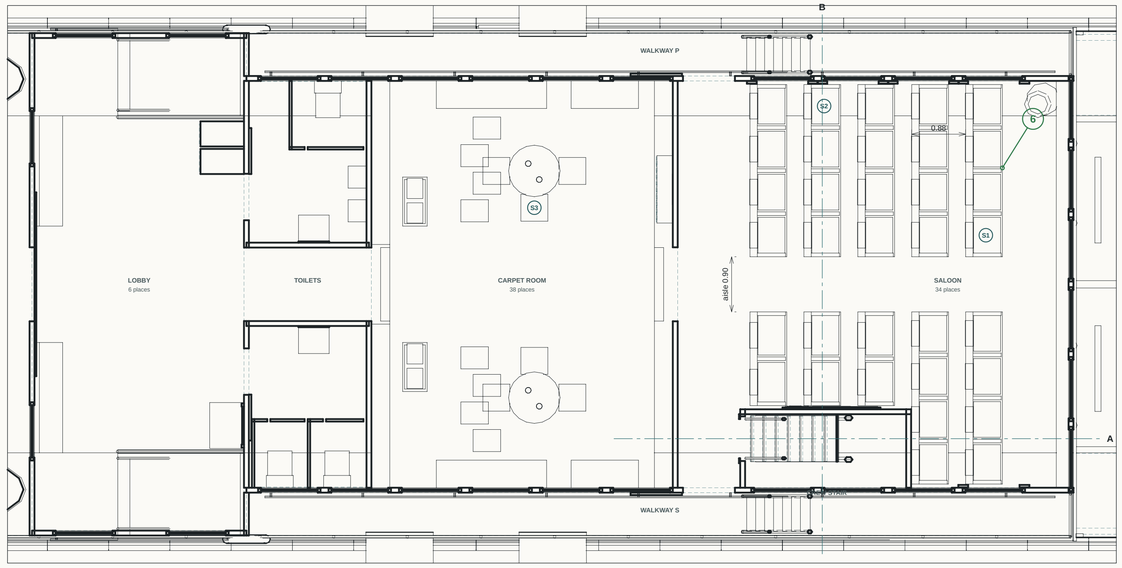
YARD	SHIP	DRAWING No.
Johansson Shipyard, Innoshima **JOHANSSON SHIPYARD**	M/V MINATO MARU hull No. 127, built 1994	**JS-127-X2**

TITLE	SCALE
ALONGSIDE KITANO-JIMA · findings and fixes **The owner's screenshot of 3 November 2026, surveyed again**	1:100; bow, detail 1:50 on A1

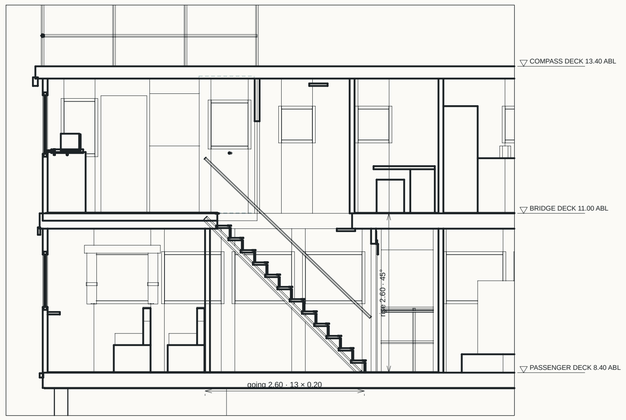
SOURCE	DATA	STATUS	SHEET
tools/ferry-exterior-cad.mjs	buildFerryModel() as built	AS-BUILT SURVEY	2 / 3

DRAWN	CHECKED	DATE	REV
Claude, AI survey	owner review open	2026-11-03	B · fixes, stair, saloon

AS-BUILT SURVEY OF THE GAME MODEL · Johansson World vessel · no flag, class or construction approval · drawn by projecting buildFerryModel() with hidden lines removed



PASSENGER DECK · plan cut 1.5 m above it SCALE 1:50



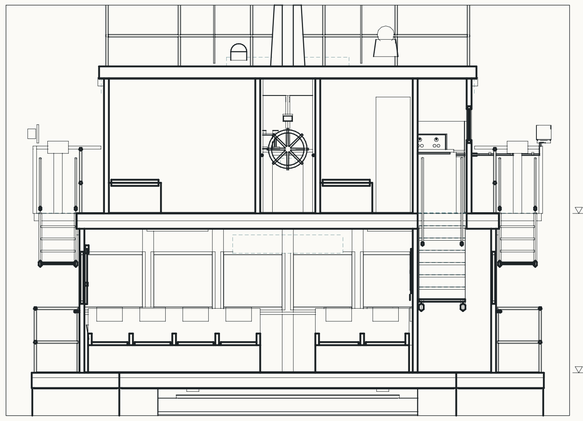
SECTION A–A · through the crew stair, looking to starboard SCALE 1:50

LINES

- Cut by the plane of the plan or section
- Visible outline beyond
- Visible feature edges
- Paint and carpet edges
- Where a section is taken
- (S) A seat anyone may take



BRIDGE DECK · plan cut 1.5 m above it SCALE 1:50



SECTION B–B · through the saloon, looking forward SCALE 1:50

WHAT CHANGED IN REV B · the owner, 3 November 2026

Crew stair to the wheelhouse
"We need an interior stair case to the bridge as well." A straight flight, 0.8 m clear, 13 risers of 0.2 m at 45°, in a trunk on the saloon's starboard side behind a door marked 乗組員専用 (crew only); its well rises through the master's cabin's corner, walled off, into the wheelhouse's aft starboard corner, railed, beside the starboard wing door (moved 0.85 m forward). The chart table moved to the aft bulkhead (finding 5); the master's bunk to the corridor's wall.

A saloon to do nothing in
"The accommodation needs to be more cosy and comfortable, it should be fun to sit there and do nothing." 34 reclining seats in terracotta moquette, 0.7 m a place, wooden arms, white head covers, five rows at 0.88 m facing the bow (finding 6), a 0.9 m aisle; a rose-brown carpet, cream curtains tied back at the side windows, a teak ledge under the bow windows, a pothos from Sakura.

The carpet room
A round low table each side with zabuton, tea cups, pillows and folded blankets; the reading rack at the forward partition: the town paper (printed at Front-Row) and manga and paperbacks from Sakura.

Seats you can take
S1 by the bow windows, S2 the port window seat, S3 a cushion in the carpet room: sit and she carries you across (game.js keeps a seat aboard in her frame).

Places for 100
Open deck 22 · lobby benches 6 (now counted) · carpet room 38 · saloon 34 = 100 passengers, as certified.

NOTES
Plans are cut 1.5 m above their deck and sections through the plane marked, both from buildFerryModel() as the game builds it; what is cut is drawn heavy, what lies beyond in outline. Rooms are named from the general arrangement (src/world/ferry-ship.js FERRY_GA).

YARD	SHIP	DRAWING No.
Johansson Shipyard, Innoshima **JOHANSSON SHIPYARD**	M/V MINATO MARU hull No. 127, built 1994	**JS-127-X3**

TITLE	SCALE
ACCOMMODATION AND CREW STAIR **Passenger deck, bridge deck and two sections from the game model**	1:50 on A1

SOURCE	DATA	STATUS	SHEET
tools/ferry-exterior-cad.mjs	buildFerryModel() as built	AS-BUILT SURVEY	3 / 3

DRAWN	CHECKED	DATE	REV
Claude, AI survey	owner review open	2026-11-03	B · fixes, stair, saloon

AS-BUILT SURVEY OF THE GAME MODEL · Johansson World vessel · no flag, class or construction approval · drawn by projecting buildFerryModel() with hidden lines removed